Amal Nambiar

amalnambiar349@gmail.com | 6238552513 | Chennai, Tamil Nadu
linkedin.com/in/amal-nambiar | github.com/Sherlock-Holmes-221B-droid

August 1, 2026
Hiring Manager
NVIDIA
Bengaluru, India

Subject: Application for Senior Software Engineer – Autonomous Driving Software

Dear Hiring Manager,

I'm applying for the Senior Software Engineer role on the Autonomous Driving Software team. As Lead Engineer at Mahindra & Mahindra, I work through the development and validation loop for AEB, Lane Keeping Assistant, Emergency Lane Keeping, and Adaptive Cruise Control primarily at the MIL, SIL, and VIL level, where I work through sensor and actuator interfaces, Vehicle IO / signal alignment, and calibration as part of closing out a feature. I've also spent time around HIL setups and run simulations there, though I'd call that exposure rather than deep hands-on ownership happy to be upfront about that.

A large part of my job is figuring out why a feature degrades in the field and tracing that back to a fault, a bad calibration, or a signal timing issue not just flagging it, but working through diagnostic message flows and factory vs. service calibration paths to actually close it out. To support this, I've built my own tooling: a CAN/MF4 signal analysis utility for log/replay debugging, a Python GUI implementing ISO 18571 correlation for CAE-to-test signal comparison, and a Euro NCAP KPI extraction tool, all built because the off-the-shelf workflow was too slow for the pace we needed during test campaigns. That KPI validation and log/replay experience maps directly onto what you're describing for feature-level debug.

Beyond validation, I work at the algorithm layer itself, tuning and modifying the perception and control logic for AEB and LKA, building sensor fusion and tracking behavior, and implementing Kalman Filter/EKF-based state estimation and search-based planning algorithms (multiple A* variants, Dijkstra, RRT) from scratch in Simulink and Python. Owning both the algorithm and its vehicle-level validation has given me a grounded sense of how perception, planning, and control actually have to come together once you move off the bench and onto the vehicle something I understand matters a great deal for this role.

My graduate research at IIT Madras was on modeling and debugging a nonlinear piezomagnetoelastic energy harvesting system, which mostly meant chasing down mismatches between simulation and physical test data the same instinct I bring to root-causing issues on the vehicle side. I've also worked under ASPICE process discipline and ISO 26262 requirements throughout my time at Mahindra, and I'm used to coordinating across functional teams system, hardware, and test to get a feature closed out.

I'd welcome the chance to bring this integration-and-algorithm background to NVIDIA's autonomous driving team. Thank you for considering my application happy to walk through any of this in more detail.

Sincerely,
Amal Nambiar
6238552513